PROBLEM 02 | Polar Kinematics

Kinematics, Spiral

Original statement · Problems.pdf, p. 6

A planar spiral is given by the equation $r = \alpha e^{\beta\theta}$, where α and β are constants, r is the radial coordinate, and θ is the angular coordinate. If a particle moves along the spiral according to the equation $\theta = \delta t^n$ (where δ is a constant), determine:

(a) The radial and transverse components of the velocity.

(b) The radial and transverse components of the acceleration.

(c) The tangential and normal components of the velocity.

(d) The tangential and normal components of the acceleration.

(e) The radius of curvature of the spiral.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle P moves along a logarithmic/exponential planar spiral $r(\theta) = \alpha e^{\beta\theta}$ with a prescribed power-law angular motion $\theta(t) = \delta t^n$.
- **Hypotheses:** $\alpha, \beta, \delta > 0$ are positive constants, $n > 0$ is an integer or real exponent, $t \geq 0$. Motion is strictly 2D planar in the Oxy plane ($z = 0$).
- **Degrees of Freedom:** $\mathrm{CDOF} = 1$ parameterized by time t.

- **Parameters & Units:** $[r] = \mathrm{m} \implies [\alpha] = \mathrm{m}$, β is dimensionless ($[-]$), $[\theta] = \mathrm{rad} \implies [\delta] = \mathrm{rad} \cdot \mathrm{s}^{-n}$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Basis:** $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Polar Basis:** $\mathcal{B}_{\text{polar}} = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$.
- **Rotation Matrix $[{}_0R_\theta]$:**

$$[{}_0R_\theta] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \\ \mathbf{k}_0 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{i}_0 \\ \mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix}$$

- **Time Derivatives of Polar Unit Vectors (Notes Sec. 2.6.2, Eqs. 2.25–2.26):**

$$\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r$$

- **Orthonormality of $[{}_0R_\theta]$:** $\det[{}_0R_\theta] = \cos^2\theta + \sin^2\theta = 1$ and $[{}_0R_\theta][{}_0R_\theta]^T = I$ (checked symbolically with sympy).
- **Intrinsic basis from the polar basis.** Since $\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta = r\dot{\theta}\,(\beta\,\mathbf{e}_r + \mathbf{e}_\theta)$ (Phase 3, part (a)), the unit tangent is $\mathbf{e}_t = (\beta\,\mathbf{e}_r + \mathbf{e}_\theta)/\quad 1+\beta^2$ and the in-plane unit normal is $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t = (-\mathbf{e}_r + \beta\,\mathbf{e}_\theta)/\quad 1+\beta^2$. Their components in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ are the columns of

$$[{}_{\text{polar}}R_{\text{tn}}] = \frac{1}{1+\beta^2}\begin{pmatrix} \beta & -1 \\ 1 & \beta \end{pmatrix}, \qquad \det = \frac{\beta^2+1}{1+\beta^2} = 1, \qquad RR^T = I$$

- This is a planar rotation by the constant angle $\psi = \arctan(1/\beta)$ (the equiangular property of the spiral). Sympy confirms that $\mathbf{a}\cdot\mathbf{e}_t$ and $\mathbf{a}\cdot\mathbf{e}_n$ reproduce the a_t and a_n of Phase 3, parts (d), with $a_n > 0$, so $\mathbf{e}_n$ points to the concave side.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Radial and Transverse Velocity Components

Pedagogical Justification: In polar coordinates, the velocity vector is given by $\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$. The radial component $v_r = \dot{r}$ measures the expansion rate, while $v_\theta = r\dot{\theta}$ measures transverse rotation.

- **Explicit Chain Rule for $\dot{\theta}(t)$ and $\dot{r}(t)$:**

$$\theta(t) = \delta t^n \implies \dot{\theta}(t) = \frac{d}{dt}(\delta t^n) = n\delta t^{n-1}$$

Using the chain rule $\dot{r} = \frac{dr}{d\theta}\dot{\theta}$:

$$\frac{dr}{d\theta} = \frac{d}{d\theta}(\alpha e^{\beta\theta}) = \alpha\beta e^{\beta\theta} = \beta r$$

$$\dot{r} = (\beta r)(n\delta t^{n-1}) = \beta n\delta t^{n-1} r = \alpha\beta n\delta t^{n-1} e^{\beta\delta t^n}$$

- **Components:**

$$v_r = \dot{r} = \beta n\delta t^{n-1} r$$

$$v_\theta = r\dot{\theta} = n\delta t^{n-1} r$$

(b) Radial and Transverse Acceleration Components

Pedagogical Justification: The acceleration in polar coordinates (Notes Eq. 2.30) accounts for the curvature of the polar grid:

$$\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

- **Second Time Derivatives (Explicit Product Rule):**

$$\ddot{\theta}(t) = \frac{d}{dt}(n\delta t^{n-1}) = n(n-1)\delta t^{n-2}$$

For $\ddot{r}(t) = \frac{d}{dt}(\beta r\dot{\theta})$:

$$\ddot{r} = \beta(\dot{r}\dot{\theta} + r\ddot{\theta}) = \beta\left((\beta r\dot{\theta})\dot{\theta} + r\ddot{\theta}\right) = \beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}$$

- **Radial Acceleration a_r:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = (\beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}) - r\dot{\theta}^2 = (\beta^2 - 1)r\dot{\theta}^2 + \beta r\ddot{\theta}$$

Substituting $\dot{\theta} = n\delta t^{n-1}$ and $\ddot{\theta} = n(n-1)\delta t^{n-2}$:

$$a_r = (\beta^2 - 1)r(n\delta t^{n-1})^2 + \beta r n(n-1)\delta t^{n-2} = n\delta t^{n-2} r\left[(\beta^2 - 1)n\delta t^n + \beta(n-1)\right]$$

- **Transverse Acceleration a_θ:**

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\ddot{\theta} + 2(\beta r\dot{\theta})\dot{\theta} = r\ddot{\theta} + 2\beta r\dot{\theta}^2$$

Substituting derivatives:

$$a_\theta = rn(n-1)\delta t^{n-2} + 2\beta r(n\delta t^{n-1})^2 = n\delta t^{n-2} r\left[(n-1) + 2\beta n\delta t^n\right]$$

(c) Tangential and Normal Velocity Components

> **Pedagogical Justification:** In the intrinsic Frenet basis, the velocity vector is strictly tangential: $\mathbf{v} = v\mathbf{e}_t = v_t\mathbf{e}_t + 0\mathbf{e}_n$. Therefore, the normal velocity component is identically zero.

- **Calculation:**

$$v_t = v = \quad v_r^2 + v_\theta^2 = \quad (\beta r\dot{\theta})^2 + (r\dot{\theta})^2 = r\dot{\theta} \quad \beta^2 + 1 = n\delta t^{n-1} r \quad 1 + \beta^2$$

$$v_n = 0$$

(d) Tangential and Normal Acceleration Components

> **Pedagogical Justification:** Tangential acceleration $a_t = \dot{v}$ measures the rate of change of speed, while normal acceleration $a_n = \frac{v^2}{\rho}$ measures directional turning.

- **Tangential Acceleration a_t (Explicit Differentiation):**

$$a_t = \frac{d}{dt}(v) = \quad 1 + \beta^2 \frac{d}{dt}(r\dot{\theta}) = \quad 1 + \beta^2(\dot{r}\dot{\theta} + r\ddot{\theta})$$

Substituting $\dot{r} = \beta r\dot{\theta}$:

$$a_t = \quad 1 + \beta^2(\beta r\dot{\theta}^2 + r\ddot{\theta}) = \quad 1 + \beta^2 n\delta t^{n-2} r\left[\beta n\delta t^n + (n-1)\right]$$

- **Normal Acceleration a_n:**

Using the Pythagorean decomposition of the total acceleration magnitude: $\|\mathbf{a}\|^2 = a_t^2 + a_n^2 = a_r^2 + a_\theta^2$:

$$a_n = \quad a_r^2 + a_\theta^2 - a_t^2$$

Alternatively, using the cross product $\mathbf{a} \times \mathbf{v} = (a_t \mathbf{e}_t + a_n \mathbf{e}_n) \times (v \mathbf{e}_t) = -v a_n \mathbf{e}_b$:

$$a_n = \frac{\|\mathbf{v} \times \mathbf{a}\|}{v} = \frac{|v_r a_\theta - v_\theta a_r|}{v}$$

Computing the numerator:

$$v_r a_\theta - v_\theta a_r = (\beta r \dot{\theta})(r\ddot{\theta} + 2\beta r \dot{\theta}^2) - (r\dot{\theta})((\beta^2 - 1) r \dot{\theta}^2 + \beta r \ddot{\theta})$$

$$= \beta r^2 \dot{\theta}\ddot{\theta} + 2\beta^2 r^2 \dot{\theta}^3 - (\beta^2 - 1) r^2 \dot{\theta}^3 - \beta r^2 \dot{\theta}\ddot{\theta} = (2\beta^2 - \beta^2 + 1) r^2 \dot{\theta}^3 = (\beta^2 + 1) r^2 \dot{\theta}^3$$

Dividing by $v = r\dot{\theta} \quad \beta^2 + 1$:

$$a_n = \frac{(\beta^2 + 1) r^2 \dot{\theta}^3}{r\dot{\theta} \quad \beta^2 + 1} = r\dot{\theta}^2 \quad \beta^2 + 1 = n^2 \delta^2 t^{2n-2} r \quad 1 + \beta^2$$

(e) Radius of Curvature of the Spiral

> **Pedagogical Justification:** The radius of curvature is obtained directly from $a_n = \frac{v^2}{\rho} \implies \rho = \frac{v^2}{a_n}$.

- **Calculation:**

$$\rho = \frac{v^2}{a_n} = \frac{(r\dot{\theta} \quad 1 + \beta^2)^2}{r\dot{\theta}^2 \quad 1 + \beta^2} = \frac{r^2 \dot{\theta}^2 (1 + \beta^2)}{r\dot{\theta}^2 \quad 1 + \beta^2} = r \quad 1 + \beta^2$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Geometric Invariance:** In a logarithmic spiral $r = \alpha e^{\beta\theta}$, the angle ψ between the radial line and the tangent vector satisfies $\tan\psi = \frac{r}{dr/d\theta} = \frac{r}{\beta r} = \frac{1}{\beta} = \text{constant}$. This is the celebrated equiangular spiral property!
- **Radius of Curvature:** $\rho = r \quad 1 + \beta^2$ is strictly proportional to current radial distance r.
- **Units Check:** $[\rho] = [r][\quad 1 + \beta^2] = \text{m} \cdot 1 = \text{m}$ (Consistent).
- **Limiting Case $\beta \to 0$:** The spiral degenerates into a pure circle $r = \alpha$, where $\rho = r \quad 1 + 0 = r$, recovering the circle radius of curvature.

Complete Step-by-Step Solution

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